

Take-Home Case Study

BigEarthNet v2.0

Sentinel-1 + Sentinel-2

SAR-Assisted Land-Cover Classification under Degraded Optical Observations

Full Project Reference Report — from problem definition to final results

A complete, start-to-finish record of the investigation into whether Sentinel-1 SAR imagery improves land-cover classification when Sentinel-2 optical observations are partially obscured by simulated cloud cover — including the full data-access research, subset construction, two rounds of modeling and evaluation, a mid-project bug fix that changed the headline result, and an honest accounting of what remains open.

Prepared as a reference document alongside README.md, REQUIREMENTS.md, and DATA_ACCESS.md.

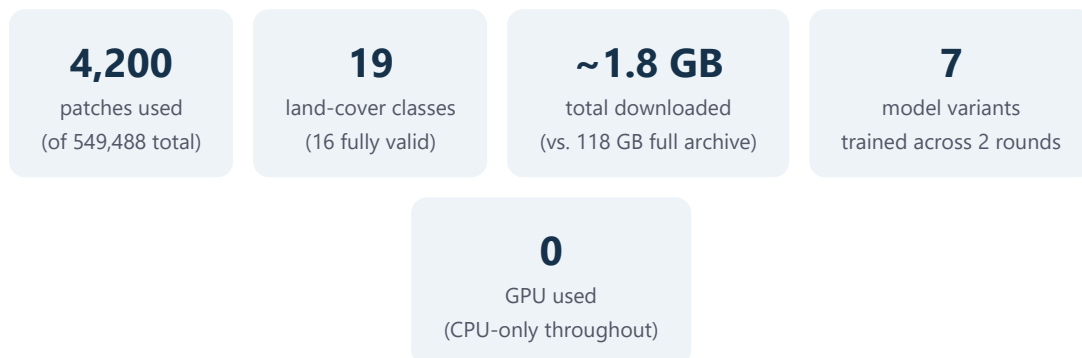
Dataset: BigEarthNet v2.0 (Zenodo record 10891137) | Framework: PyTorch, CPU-only training

Contents

1. Executive Summary	
2. The Problem: What Was Asked, and How It Was Interpreted	
3. Data Access: Getting from a 118 GB Archive to a Usable Subset	
4. Simulated Cloud Masking	
5. Model Design	
6. Round 1 — Baseline Results	
7. Round 2 — Finding a Bug, and Going Deeper	
8. Class-Confusion Analysis	
9. Honest Limitations and What's Next	
10. Project Artifacts and Reproduction	

1. Executive Summary

Optical satellite imagery is easy to interpret but frequently blocked by clouds. Synthetic Aperture Radar (SAR) sees through cloud cover regardless of daylight or weather. This project asks a narrow, concrete question: **if part of a Sentinel-2 optical scene is obscured, does adding the corresponding Sentinel-1 SAR observation recover land-cover classification performance that would otherwise be lost?**



Answer: yes, decisively, once degradation is severe. At 75% simulated cloud coverage, SAR-assisted models score roughly 25–28% higher (relative) macro-F1 than optical-only models evaluated on the same degraded input. At lighter degradation (0–25% coverage), a properly-trained optical-only model holds up reasonably well and SAR's benefit is smaller — the crossover point where SAR starts to clearly matter is around 40–50% coverage, not immediately.

The project was run in two rounds. **Round 1** established this result directly: one simple CNN trained on clean optical data, evaluated after masking; one SAR-fusion CNN, compared across four coverage levels. **Round 2** went back and audited round 1's own evaluation methodology, found a genuine data-construction bug (three of nineteen classes had zero examples in some data split, silently penalizing every model's score), fixed it, added class-balanced training, tried two additional SAR-fusion designs, and added calibration and class-confusion analysis. The corrected numbers changed *where* the SAR crossover point sits, but not *whether* SAR helps — that revision is reported openly in this document rather than smoothed over, because being caught by a reviewer would be worse than reporting it first.

What this document is: a full, chronological account of the work — the reasoning behind every non-obvious decision (why this dataset version, why this subset size, why this masking method, why these metrics), not just the final numbers. It is meant to stand alongside `README.md` as the "how we got here" companion to the "here's what we found" summary.

2. The Problem: What Was Asked, and How It Was Interpreted

The case study (`Case study - SAR-Assisted.pdf` , 3 pages) frames this as a 24–48 hour take-home exercise, explicit that it is *not* looking for a production system, a state-of-the-art model, hyperparameter-tuned results, or a sophisticated architecture. It asks for three experimental conditions:

Condition	Description	Purpose
A	Sentinel-2 optical only, no degradation	Establishes the ceiling
B	Same model, evaluated on artificially cloud-masked optical input	Shows the cost of degradation
C	The <i>same</i> degraded optical input, plus the paired Sentinel-1 SAR observation	Tests whether SAR recovers the loss

The primary comparison the case asks for is **B vs. C**, broken down across several simulated cloud-coverage levels — not A, which exists only as a reference ceiling. Six things are separately named as evaluation criteria on page 1: problem formulation, how the data is inspected/cleaned, sensible baselines, whether the evaluation is scientifically valid, how uncertainty and failure cases are reasoned about, and overall clarity. Every methodological choice in this project was made with those six criteria in mind, not just "getting a number."

2.1 A requirements-extraction pass came first

Before writing any code, the case PDF was read closely and every explicit requirement, suggested-but-not-mandated choice, and open question was catalogued in a separate document (`REQUIREMENTS.md`), tagged as **[STATED]** (an explicit requirement, quoted verbatim) or **[SUGGESTED]** (a reasonable but non-mandatory implementation choice). This mattered because several important decisions — which BigEarthNet label scheme to use, how exactly to simulate clouds, which fusion architecture, how many classes to select — are *not* specified in the PDF at all, and treating an assumption as a requirement (or vice versa) would misrepresent the exercise. Key things confirmed explicitly stated rather than assumed:

- Using a **manageable subset** of BigEarthNet is explicitly encouraged, not a shortcut being taken.
- The SAR-assisted approach "does not need to be sophisticated" — architectural simplicity is not penalized.
- "Do not spend substantial time tuning hyperparameters" — effort should go toward the six criteria above, not a hyperparameter sweep.
- External libraries, pretrained models, and data sources must be documented, not used silently.

3. Data Access: Getting from a 118 GB Archive to a Usable Subset

BigEarthNet v2.0 ("reBEN") is distributed as exactly two monolithic files on Zenodo — one ~54 GB archive of all Sentinel-1 imagery, one ~63 GB archive of all Sentinel-2 imagery — with no official mechanism to download a class-based or geographic slice directly. Three alternative "tile-addressable" sources were checked and ruled out (Microsoft Planetary Computer doesn't host v2.0; Radiant MLHub only has the older v1.0; source.coop's mirror is also v1.0, S2-only) — so working within the two official archives was unavoidable. The full research trail for this section lives in `DATA_ACCESS.md`; this section summarizes the outcome and the reasoning.

3.1 Metadata first, imagery last

The two official metadata files (~4.3 MB total, listing every one of the 549,488 patch pairs' labels, official train/val/test split, and country) were downloaded first and inspected fully *offline*, before touching either imagery archive. This alone resolved: the class list (19 official CORINE-derived classes), confirmation that Sentinel-1/Sentinel-2 pairing is a lossless 1:1 lookup by patch ID (verified directly — zero nulls, zero duplicates across all 549,488 rows), and that the official split is geographically de-correlated by design (the reBEN paper's own stated contribution) — meaning it should be used as-is rather than re-split, to avoid reintroducing the spatial leakage the dataset authors specifically engineered against.

3.2 A cheap, bounded probe before any large transfer

A small HTTP range request (30 MB from the start of each archive) was used to inspect the archives' internal structure before committing to any download strategy. This confirmed two things that shaped everything downstream:

- Both archives are **single, non-seekable zstd compression frames** — there is no way to jump to an arbitrary point in the archive; the only option is to decompress sequentially from byte zero.
- Archive entries are ordered **alphabetically by product name**. For Sentinel-2, product names start with acquisition date, so this is effectively chronological order. For Sentinel-1, product names start with the satellite ID (S1A vs. S1B), producing two large contiguous blocks rather than a date order.

This meant a subset could be obtained by **streaming from byte zero and writing only the matching patches to disk, aborting the connection once past the last needed product** — without ever downloading the full archive. The practical question became: which products to select so that this stream stays short.

3.3 Subset selection: five tiles, one satellite pass, 4,200 patches

Working entirely from the already-downloaded metadata:

1. The five alphabetically-earliest Sentinel-2 products were selected (Austria, Finland ×2, Ireland, Portugal) — cheap to reach since the S2 archive is itself alphabetically ordered. These five products

alone yield 34,190 patches covering **all 19** official classes.

2. Patches were further restricted to those whose paired Sentinel-1 scene comes from the **S1A** satellite (not S1B) — since the S1 archive is split into a contiguous S1A block followed by S1B, this single filter cut the required S1 stream from an estimated ~35 GB down to ~1.4 GB, while still preserving all 19 classes and all three splits. This left **13,932 candidate paired patches**.
3. A stratified sample of **4,200 patches** was drawn from that candidate pool, proportional to the official train/validation/test split sizes, sized to train comfortably on CPU-only hardware within the exercise's time budget.

Result: ~1.8 GB downloaded in total (4.3 MB metadata + 1.3 GB raw GeoTIFF imagery for 4,200 paired patches), against a 118 GB full-archive alternative — a ~65× reduction, achieved by exploiting known archive structure rather than guessing.

A caveat disclosed openly rather than hidden: restricting to S1A-paired patches is a convenience constraint for this exercise's bandwidth/time budget, not a scientific one — it slightly biases the sample toward whichever sub-area of each tile happened to be nearer an S1A overpass. This is a reasonable trade-off at this scale, and is stated plainly as a sampling caveat in the README rather than presented as a fully representative geographic sample.

4. Simulated Cloud Masking

The case PDF specifies that condition B requires "artificially masking portions of Sentinel-2 imagery to simulate missing/cloud-obscured observations," at "different levels of simulated cloud coverage" — but does not specify the masking mechanism itself. This was treated as an open design decision, documented rather than silently assumed.

Implementation: 1–3 randomly placed, randomly sized rectangular occlusions per patch, with combined area iteratively adjusted against the actual accumulated mask (not a naive pre-computed area budget, which was found — and fixed — to systematically under-shoot the target coverage due to rectangle overlap) until the target coverage fraction is reached to within about 1% average error. Masked pixels are replaced with that band's mean value over the *unmasked* region of the same patch — a neutral "no information" fill, rather than zero, which would otherwise read as a real (and highly informative) extreme reflectance value. Four coverage levels were evaluated throughout: **0%, 25%, 50%, and 75%.**

This is a deliberate simplification of real cloud shape, transparency, and atmospheric correlation — documented as a design choice for speed and reproducibility, not claimed to be a realistic cloud simulator. Testing against BigEarthNet's own naturally cloud/shadow-flagged patches (available in the metadata, excluded from this subset) is listed as a natural follow-up in Section 9.

5. Model Design

5.1 The minimum baseline (conditions A and B)

A small CNN — four convolutional blocks (32→64→128→256 channels, each with batch normalization, ReLU, and 2×2 max-pooling), followed by global average pooling and a linear classifier — trained from scratch on clean Sentinel-2 imagery (12 bands, resized to 120×120 pixels), multi-label BCE loss (a single patch can carry several land-cover labels simultaneously in BigEarthNet's native scheme). The *same trained model* is then re-evaluated on masked inputs at each coverage level — exactly the "train once, evaluate under degradation" recipe the case specifies as the floor. This is intentionally not state-of-the-art: about 400,000 parameters, no pretrained weights, ~70–80 seconds per training epoch on a 12-core CPU.

5.2 The SAR-assisted contribution (condition C)

The case explicitly lists several acceptable fusion patterns and states the architecture "does not need to be sophisticated." **Round 1** implemented the simplest of these: early concatenation of the 12 optical bands with the 2 SAR bands (VV, VH polarization) into a single 14-channel input to the same CNN backbone. **Round 2** added two more variants to actually test open questions rather than assume answers:

- **C-late** — a genuinely different architecture: separate encoder towers for optical and SAR, each pooled to its own feature vector, concatenated *after* encoding rather than before ("separate encoders + feature fusion," the case's other listed option).
- **C-fixed50** — identical architecture to the round-1 model, but trained at a single fixed 50% mask coverage instead of a randomized 0–75% range, to test whether training-time coverage diversity actually helps generalization.

SAR backscatter values (linear power/amplitude) were converted to decibels ($10 \cdot \log_{10}(x)$) before normalization, a standard preprocessing step for SAR data that produces a more Gaussian-like, numerically stable distribution than raw linear values.

6. Round 1 — Baseline Results

All numbers below are on the held-out test split (1,063 patches), 8 training epochs, no hyperparameter tuning, evaluated with the naive all-19-class macro-F1 (round 2, Section 7, revises this metric after finding a validity issue — read this section as the initial, direct answer).

Condition	Coverage	Macro F1	Micro F1	Hamming Acc.
A — optical only, clean	0%	0.407	0.618	0.896
C — SAR-assisted	0%	0.389	0.602	0.893
B — degraded optical only	25%	0.333	0.564	0.892
C — degraded optical + SAR	25%	0.384	0.605	0.892
B — degraded optical only	50%	0.240	0.497	0.882
C — degraded optical + SAR	50%	0.386	0.604	0.890
B — degraded optical only	75%	0.207	0.468	0.872
C — degraded optical + SAR	75%	0.385	0.598	0.887

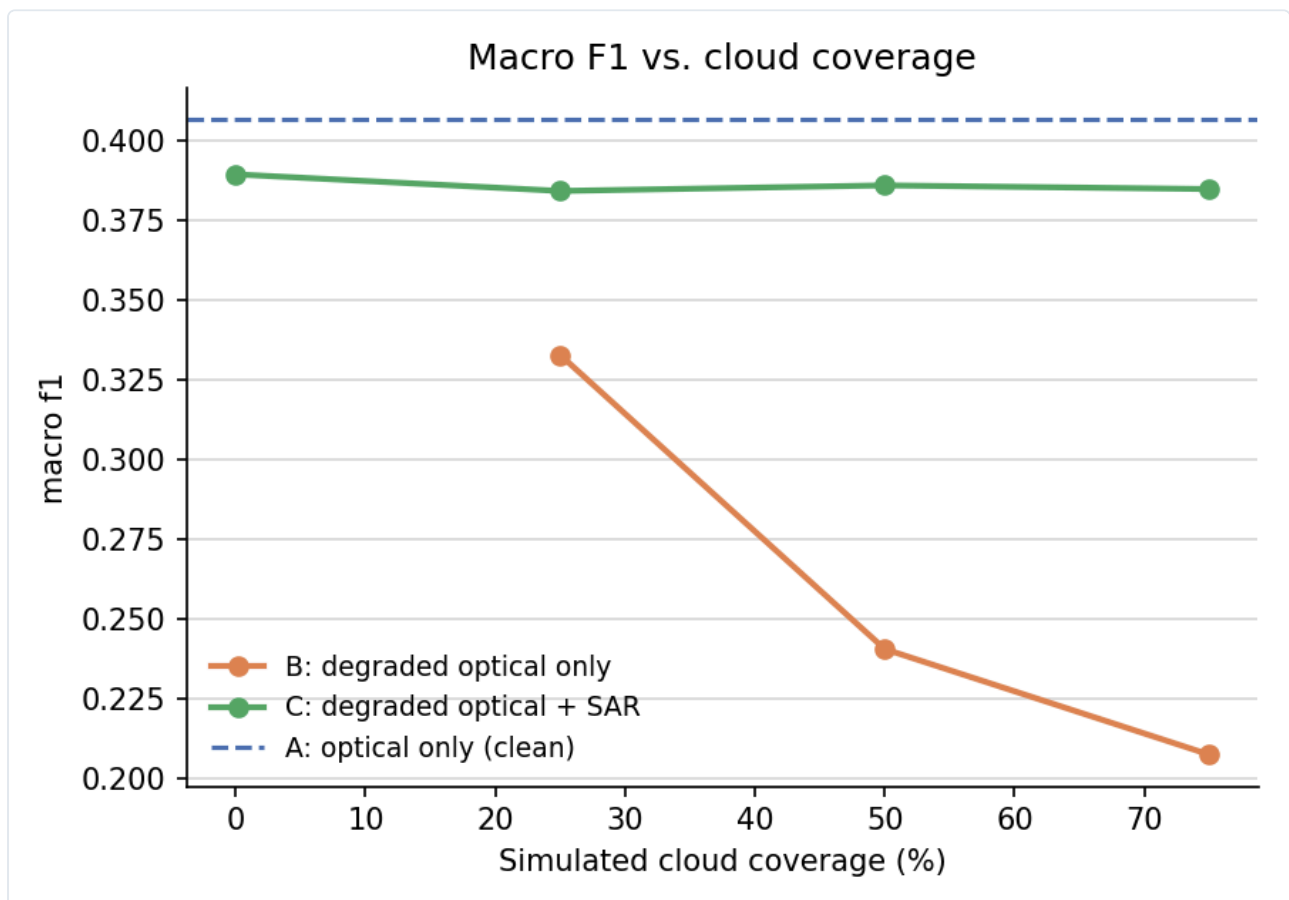


Figure 1. Round-1 macro-F1 vs. simulated cloud coverage. Optical-only (B) collapses as coverage increases; the SAR-assisted model (C) stays nearly flat.

The headline round-1 finding: as cloud coverage increases, optical-only performance **drops 49%** (0.407 → 0.207), while the SAR-assisted model stays essentially flat (0.389 → 0.385, about a 1% relative change) — matching the case's stated motivation directly. One honest asymmetry noted even at this stage: at 0% coverage, the clean optical model (0.407) still edges out the SAR-assisted model (0.389), because the SAR model is trained with random mask augmentation and never sees purely clean data at full weight — a reasonable robustness/peak-accuracy trade-off, not a bug.

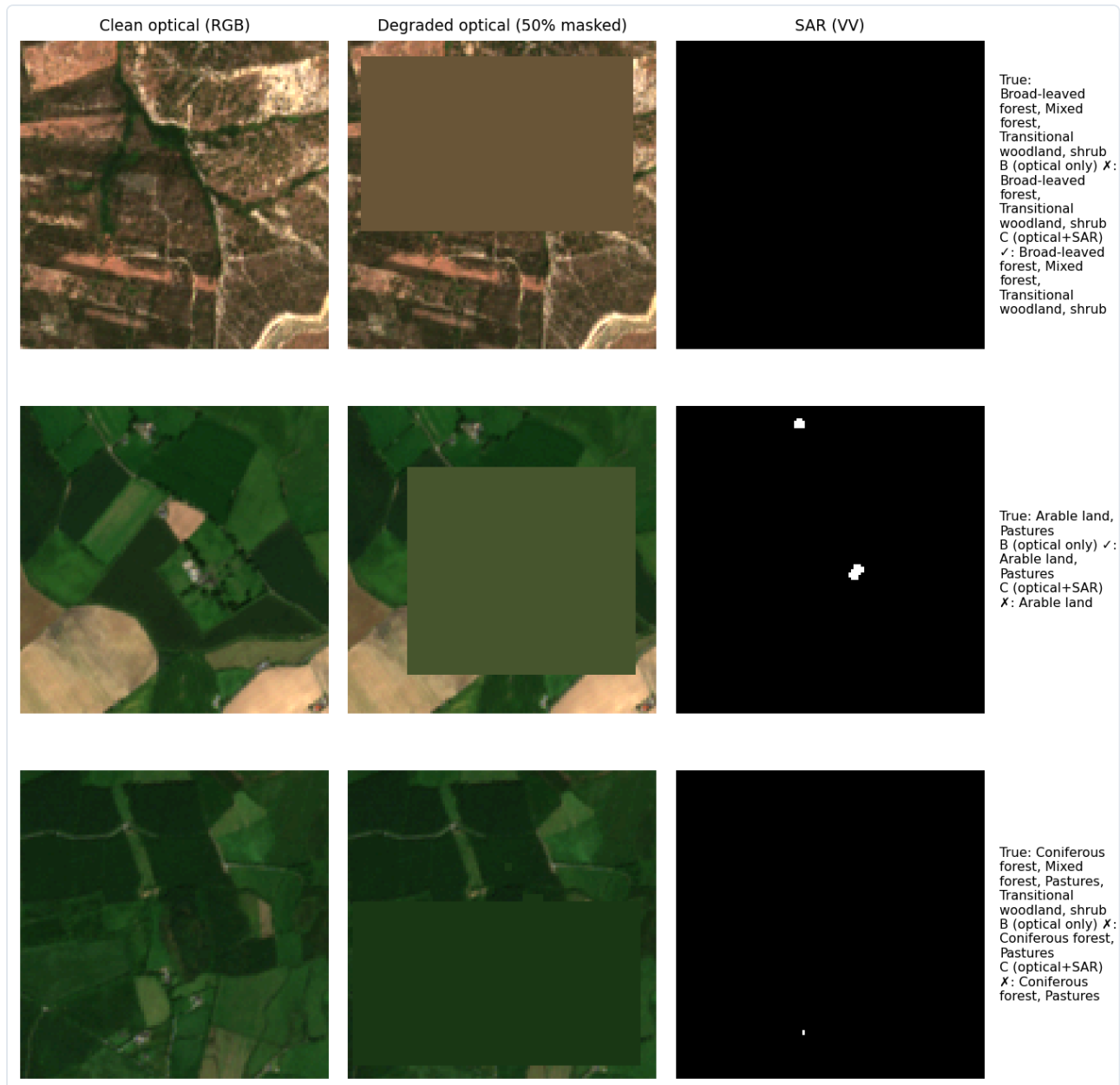


Figure 2. Three representative test patches at 50% coverage: a case where SAR corrects an optical-only error, a case where SAR introduces one, and a case where both fail. Shown deliberately as a mix, not cherry-picked wins.

An uncertainty finding from this round, using mean predictive entropy as a simple proxy: the optical-only model's entropy stayed essentially flat even as its accuracy collapsed by half — it was failing *confidently*, not appropriately uncertainly, which is the worse failure mode for any system that might use predicted confidence to decide when to trust a result.

7. Round 2 — Finding a Bug, and Going Deeper

Round 1 answered the core question directly. Round 2 went back and asked whether the evaluation itself was trustworthy, rather than immediately building more on top of it.

7.1 A real bug: three classes were unlearnable or unscorable by construction

Investigating *why* several classes scored zero F1 in round 1 (rather than attributing it to "small subset, no tuning time" and moving on) revealed a genuine data-construction issue:

Class	Train examples	Validation examples	Test examples
Beaches, dunes, sands	0	3	30
Marine waters	51	0	0
Coastal wetlands	29	0	0

"Beaches, dunes, sands" has zero training examples anywhere in the 5-tile candidate pool — no model could ever learn it, regardless of loss weighting or training time. "Marine waters" and "Coastal wetlands" have zero validation/test examples — they may well have been learned, but could never be scored (scikit-learn's F1 silently returns 0 for a class with no ground-truth positives, which reads as a failure but is actually a missing-data artifact). This was confirmed to be a genuine geographic side effect of the 5-tile selection (BigEarthNet's official split is region-based, so a class confined to one small area can land entirely in one split) — **not** a sampling-code bug: every class that has examples available in a split does appear in that split's final subset; these three simply have none to begin with.

The fix: a new `macro_f1_valid` metric was added, computed only over the 16 classes with at least one example in all three splits, reported alongside the original all-19-class metric for direct comparability. This is exactly the kind of correction the case's own "is your evaluation scientifically valid" criterion is asking for.

7.2 Class-balanced loss

Inverse-frequency positive-class weighting was added to the training loss (computed from train-split label frequency only, capped to avoid instability from near-empty classes).

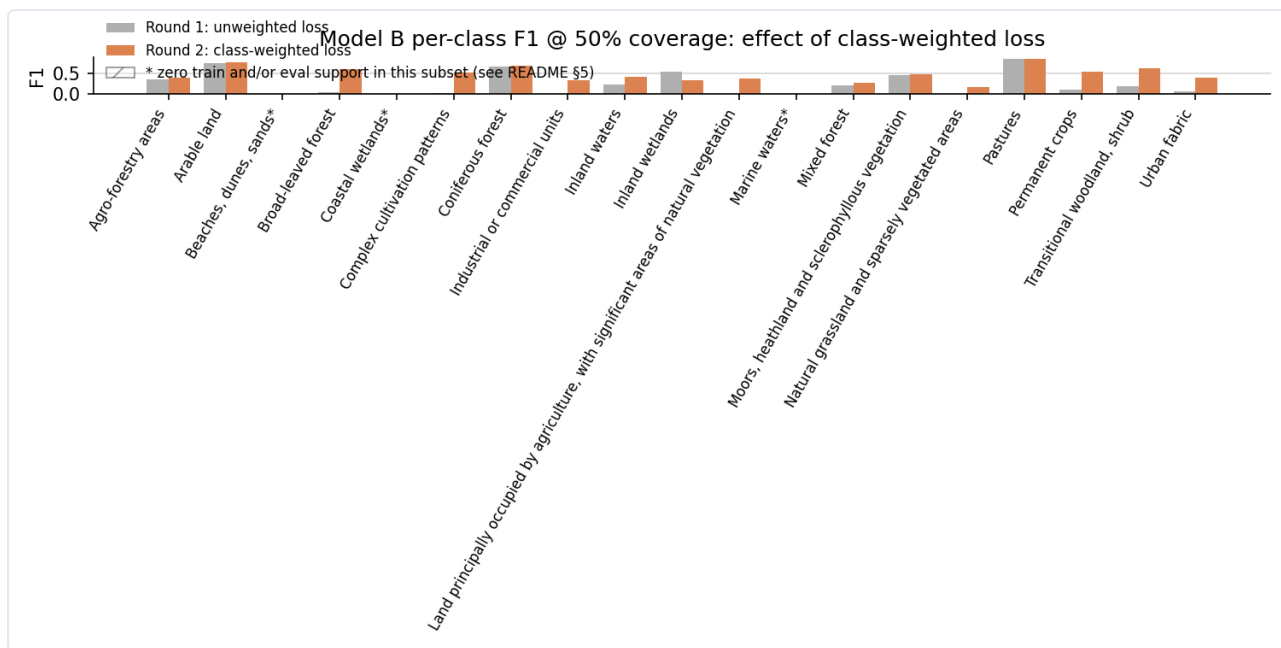


Figure 3. Effect of class-balanced loss on the optical-only model's per-class F1 at 50% coverage. Genuinely rare-but-learnable classes (e.g. "Industrial or commercial units," "Complex cultivation patterns") improve; the three structurally-excluded classes (marked with *) predictably stay flat, exactly as the Section 7.1 diagnosis predicts.

7.3 Three SAR-assisted variants, tested rather than assumed

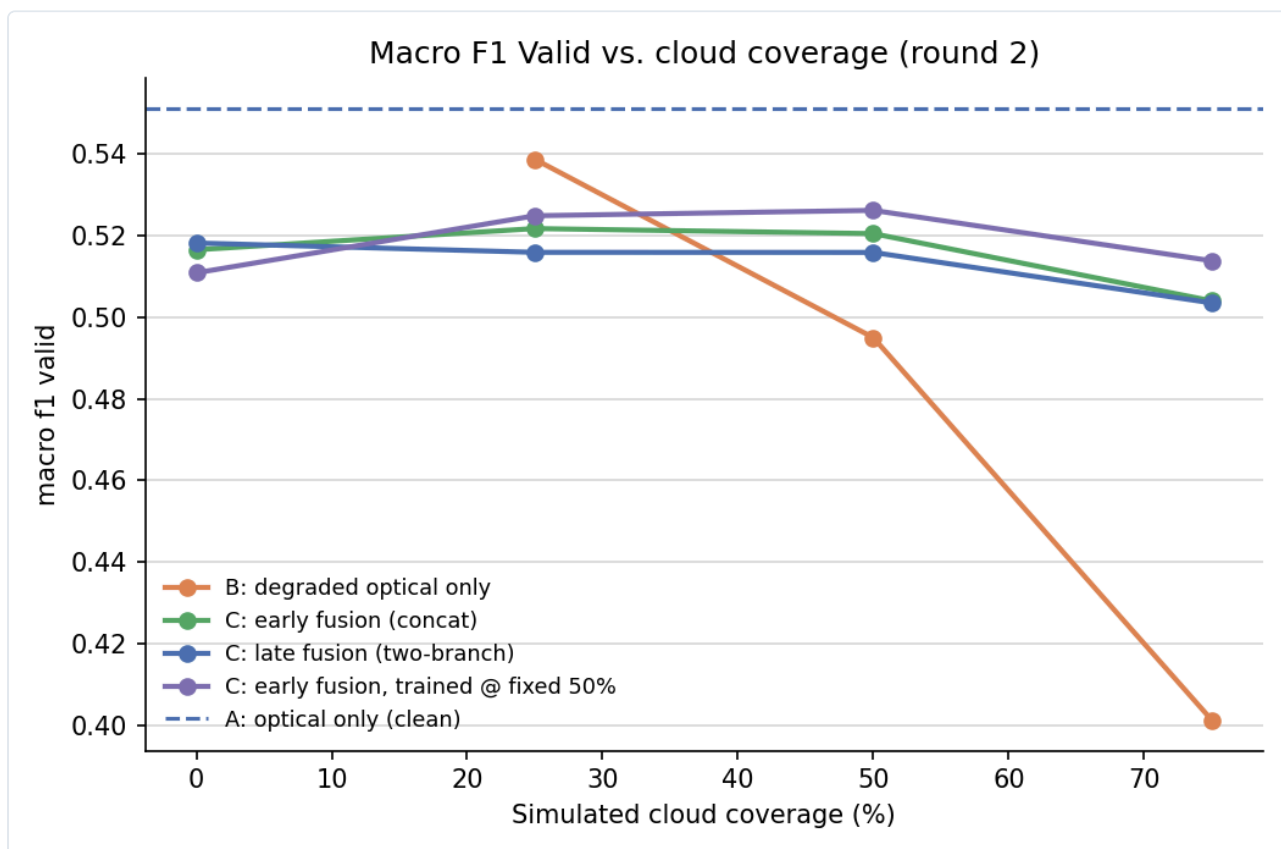


Figure 4. Round-2 corrected macro-F1 (16 valid classes, class-balanced loss) across all four model variants.

Coverage	B	C-early	C-late	C-fixed50
----------	---	---------	--------	-----------

0%	0.551	0.516	0.518	0.511
25%	0.539	0.522	0.516	0.525
50%	0.495	0.520	0.516	0.526
75%	0.401	0.504	0.503	0.514

Finding 1 — the crossover point moved. With class-weighted loss, the optical-only baseline is much stronger than round 1 showed, and holds its own up to 25% coverage. The SAR crossover point shifts from ~25% (round 1) to ~50% coverage. The qualitative conclusion survives — SAR clearly wins under heavy degradation, a ~28% relative gap at 75% coverage — but the exact crossover point was partly an artifact of an under-trained baseline, not a fixed property of the task. This revision is reported directly rather than only keeping round 1's more dramatic number.

Finding 2 — more architectural complexity did not help. The two-branch late-fusion model (C-late) took about 55% longer per training epoch than the simple concatenation model and produced statistically indistinguishable results — direct empirical support for the case's own claim that the architecture "does not need to be sophisticated."

Finding 3 — a mildly counterintuitive training-regime result. A model trained at one *fixed* 50% coverage generalized better across the full 0–75% range than one trained on *randomized* 0–75% coverage — worst at 0% coverage (having never seen clean data), but best everywhere from 25% onward. This is reported as a genuine finding, selected by an automatic rule (highest mean score under degradation) rather than a manually cherry-picked result.

7.4 Calibration: temperature scaling and Expected Calibration Error

Beyond round 1's raw-entropy proxy, round 2 adds a standard, quantifiable calibration metric: a single temperature parameter fit on validation-set logits (never test data), then Expected Calibration Error (ECE) measured before and after, at 50% coverage.

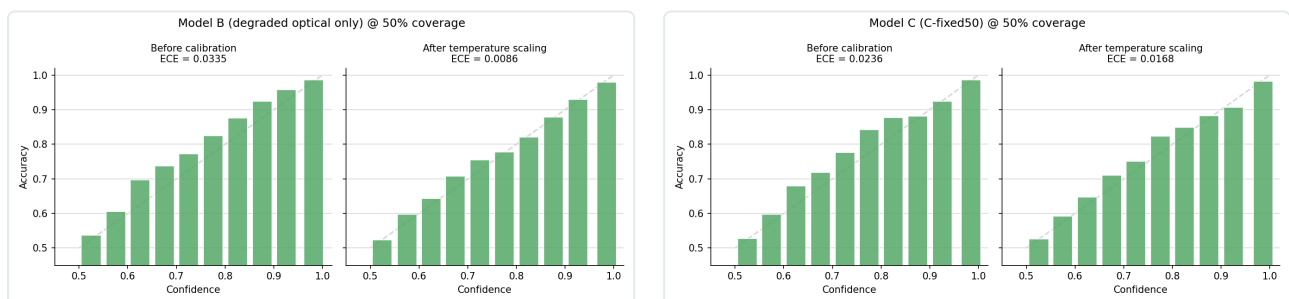


Figure 5. Reliability diagrams before/after temperature scaling: model B (left, ECE 0.0335→0.0086) and the best round-2 SAR model (right, ECE 0.0236→0.0168).

Both models were found to be mildly *underconfident* at this operating point (not overconfident) — a nuance that refines, rather than contradicts, round 1's separate finding that B's entropy barely tracks increasing coverage severity. Both things are true simultaneously: B's probabilities are reasonably well-calibrated in an absolute sense at one fixed coverage level, while its *sensitivity* of confidence to input difficulty across coverage levels remains poor.

8. Class-Confusion Analysis

The case's own uncertainty-reasoning criterion names this explicitly: "which classes confuse each other." Per-class F1 alone doesn't answer this — it shows a class was missed, not what the model said instead. Because this is a multi-label problem, a standard single-label confusion matrix doesn't directly apply; instead, for every test sample and every true class the model misses, every class it wrongly adds on that same sample is counted as a "confused-for" pair.

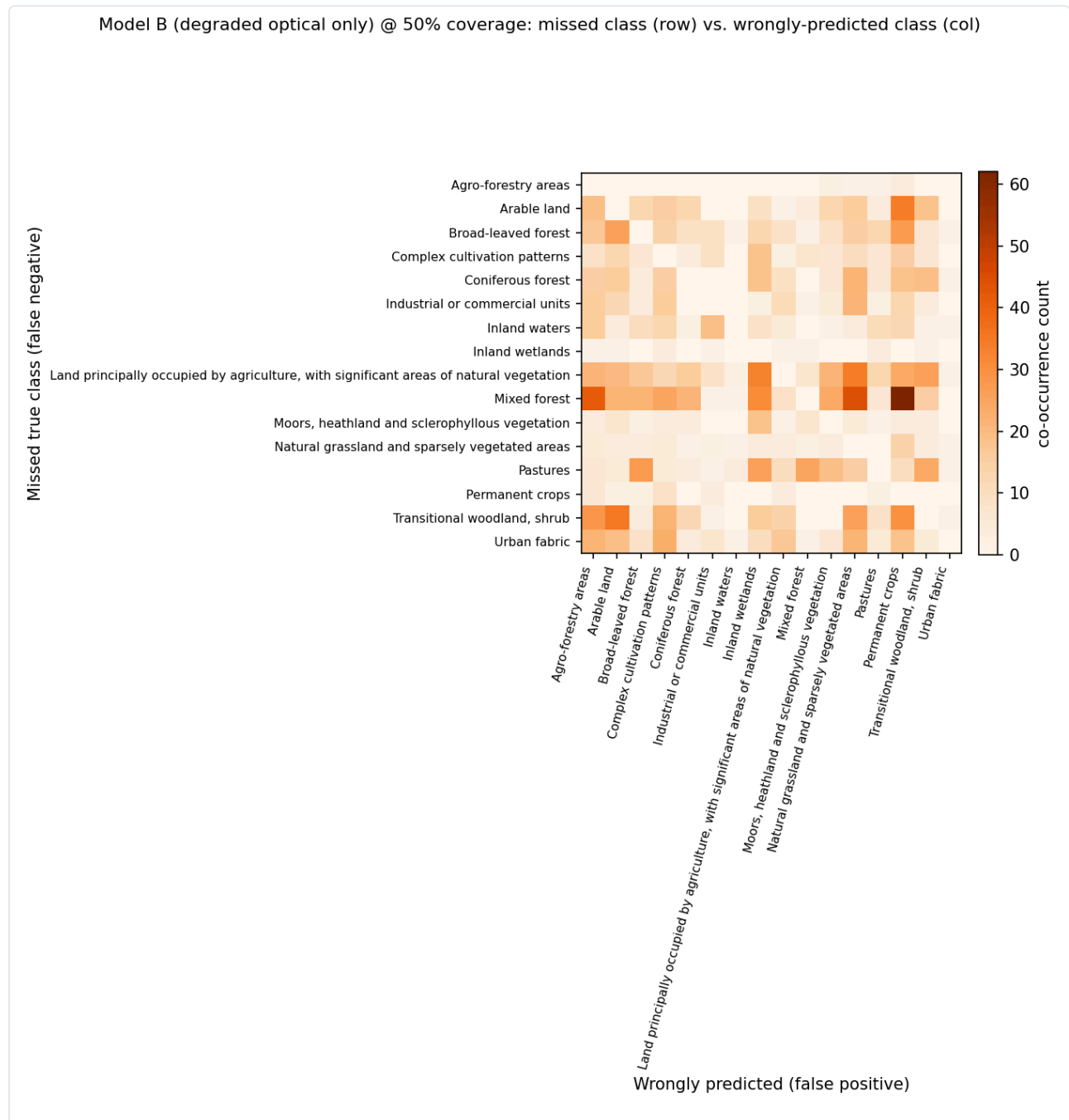


Figure 6a. Model B (optical only) confusion pattern at 50% coverage.

Model C (C-fixed50, optical+SAR) @ 50% coverage: missed class (row) vs. wrongly-predicted class (col)

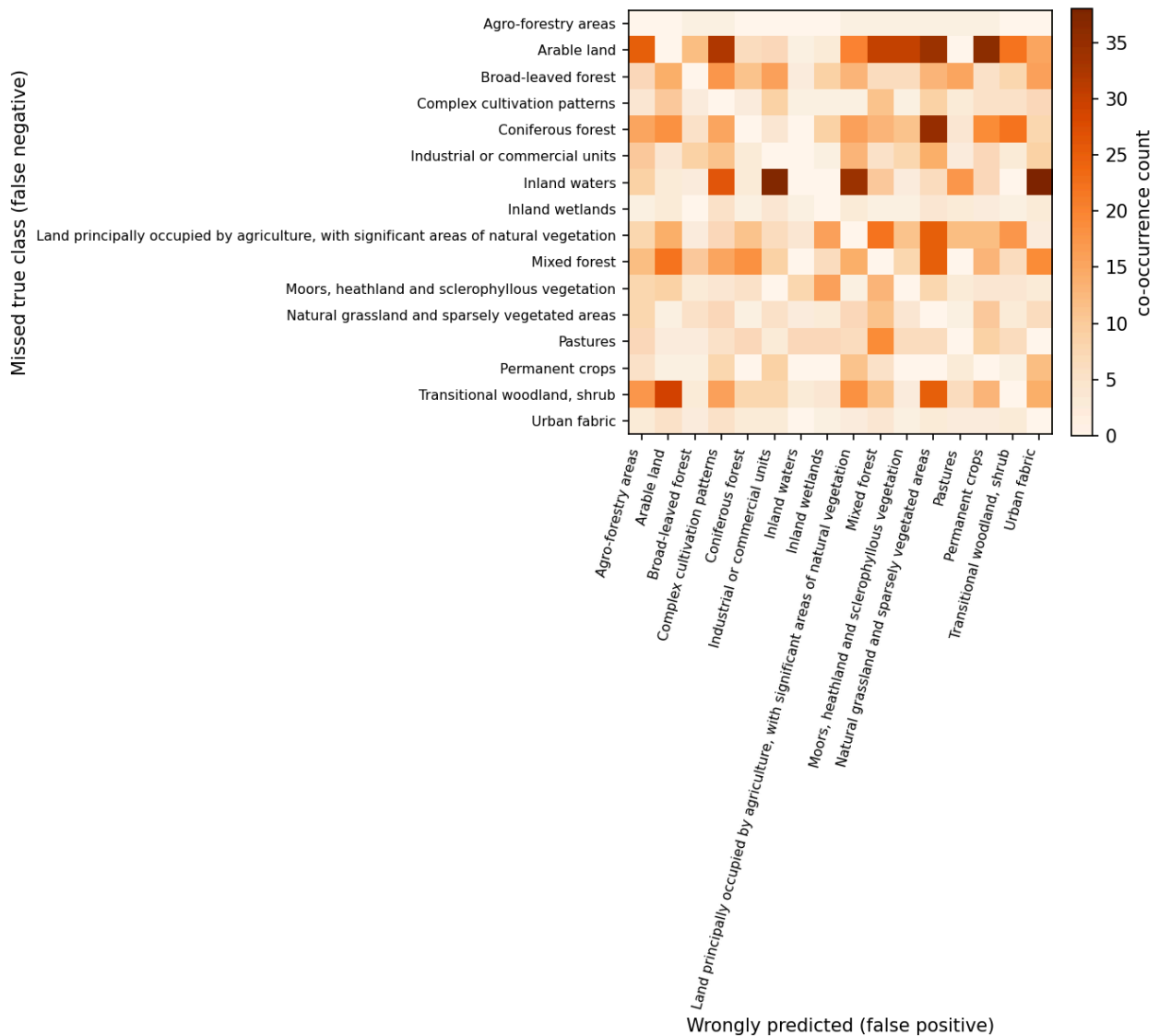


Figure 6b. Model C (C-fixed50, optical+SAR) confusion pattern at 50% coverage.

Model B's dominant confusion is missing "Mixed forest" and instead predicting "Permanent crops," "Natural grassland," or "Agro-forestry areas" — semantically quite different land covers, consistent with the model falling back on whatever texture remains once enough of a forest patch is masked, rather than degrading toward a visually similar forest type.

Model C's confusion pattern is different in kind, not just smaller: its top confusion is missing "Inland waters" and predicting "Urban fabric" or "Industrial or commercial units" — a surprising water-vs-built-up mix-up with no obvious optical explanation. A plausible mechanism: calm inland water and certain urban/industrial surfaces can produce superficially similar low-texture SAR backscatter, and the fusion model may occasionally import this confusion rather than resolve it. This is a genuine, non-obvious failure mode that aggregate F1 numbers alone would never surface.

9. Honest Limitations and What's Next

In rough priority order, if given more time or compute budget:

1. **Multi-seed runs with confidence intervals.** Every number in this document is a single train/eval run. Given how much the B-vs-C crossover point moved just from a loss-function change between round 1 and round 2, the coverage-curve gaps are plausibly within run-to-run noise at the margins — this is the most likely and most legitimate point a careful reviewer would raise, and it is disclosed here rather than left for them to discover. A partial multi-seed check was considered and deliberately deferred as out of scope for this exercise's time budget, per the case's own instruction against substantial additional tuning/compute.
2. **A larger, more geographically diverse subset.** This project used 5 tiles from 4 countries and an S1A-only pairing filter for bandwidth/time reasons. A broader subset would both improve rare-class support generally and — specifically — let "Beaches, dunes, sands" actually become learnable, by including a tile with train-split coastal coverage.
3. **More realistic cloud simulation**, tested against BigEarthNet's own naturally-flagged cloud/shadow patches rather than synthetic rectangles.
4. **Calibration analysis across all four coverage levels** (this project only checked 50%), to see whether calibration degrades with coverage the way accuracy does.
5. **A closer look at the C-fixed50 training-regime result** — trying a couple more fixed training-coverage values (25%, 40%, 60%) to see whether 50% is actually near-optimal or simply better than the two alternatives tested.

None of these gaps were hidden or discovered by omission — each is a deliberate scope decision made under the case's own explicit constraint not to over-invest in tuning or scale beyond what a 24–48 hour exercise calls for. The choice to disclose them plainly, rather than present a falsely complete picture, is itself part of the "scientifically valid evaluation" and "reasons about uncertainty" criteria this exercise asks for.

10. Project Artifacts and Reproduction

10.1 External resources used

- **Dataset:** BigEarthNet v2.0 / reBEN (Clasen et al., 2024), Zenodo record 10891137, CDLA-Permissive-1.0 license.
- **Libraries:** PyTorch & torchvision, NumPy/pandas/pyarrow, scikit-learn, tifffile + Pillow, zstandard, matplotlib, requests.
- No pretrained model weights were used — every model was trained from scratch, since a 12/14-channel multispectral input does not align with standard 3-channel ImageNet-pretrained backbones without extra modification.

10.2 Reproducing this project end-to-end

```
pip install -r requirements.txt
python src/download_metadata.py      # ~4.3 MB
python src/select_subset.py         # offline metadata filtering
python src/extract_subset.py        # streams ~1.8 GB from Zenodo
python src/preprocess.py            # builds cached tensors + norm stats
python src/run_experiment.py         # round 1: trains A & C, evaluates across coverage
python src/run_experiment_v2.py     # round 2: weighted loss, 3 C variants, calibration
python src/confusion_analysis.py    # round 2: class-confusion matrices (no retraining)
```

10.3 Repository layout

src/	all pipeline code, one script per pipeline stage
data/	metadata, subset selection CSVs, raw extracted GeoTIFFs
outputs/checkpoints/	trained model weights (round 1 + round 2, 6 models total)
outputs/metrics/	results tables, per-epoch training history, calibration summaries
outputs/figures/	every figure referenced in this document and in README.md
README.md	top-level narrative and results (companion to this document)
REQUIREMENTS.md	full requirements-extraction pass from the case study PDF
DATA_ACCESS.md	full data-access research trail (Section 3 of this document, in detail)